

PlantVillage (Bell pepper, Potato, Tomato)

Zhao *et al.* (2022) developed **RIC-Net**, a CNN that explicitly fuses Inception modules with residual (skip) connections and attention (CBAM) for leaf disease classification ¹. Their model was trained on PlantVillage data (the standard 14-plant, 38-category dataset ²) focusing on potato and tomato leaves (plus corn), and achieved **99.55%** overall accuracy on the three-disease task ¹. In short, this recent CEIA paper shows that an Inception-based network with embedded residual blocks can accurately classify PlantVillage images (e.g. potato late blight, tomato bacterial spot) with very high accuracy ¹ ².

Wheat disease dataset (leaf rust, powdery mildew)

Fang *et al.* (2023, *Appl. Sci.*) proposed a **lightweight multiscale CNN** that combines *residual blocks* (skip connections) with *Inception modules* (plus CBAM/ECA attention) to detect six wheat leaf diseases. By integrating residual “skip” connections into Inception-style convolutions, their network extracts multi-scale features while remaining compact ³. The model attained **98.7%** test accuracy on a mixed wheat disease dataset (including rust and mildew), outperforming standard CNNs (AlexNet, VGG16, InceptionResNetV2, etc.) ³. This work (Open Access) clearly matches the criteria: it is very recent (2023) and explicitly uses both Inception and residual (skip) components in the CNN architecture ³.

CottonPlant Disease dataset

Several recent studies on cotton use Inception-based models. For example, Bishshash *et al.* (2024, *Data in Brief*) released a new cotton leaf disease dataset and reported that an **InceptionV3** CNN achieved 96.03% accuracy on it ⁴. Likewise, Islam *et al.* (2023) fine-tuned various CNNs on cotton disease images and noted that *InceptionV3* (and especially *Xception*) performed strongly (*Xception* reached 98.7% accuracy) ⁵. These works show that inception architectures remain competitive for the “CottonPlant” dataset. (An additional recent model, COLD-12 ⁶, even surpassed InceptionResNetV2, though it primarily uses atrous convolutions and attention rather than a pure Inception module.) In summary, recent cotton disease papers have employed Inception-family CNNs (often with skip connections via residual/InceptionResNet components), confirming the relevance of this approach for the CottonPlant dataset ⁴ ⁵.

Sources: Zhao *et al.* (2022, *Comput. Electron. Agric.*) ¹ ²; Fang *et al.* (2023, *Appl. Sci.*) ³; Bishshash *et al.* (2024, *Data Brief*) ⁴; Islam *et al.* (2023, *Sci. Reports*) ⁵; Sharker Nirob *et al.* (2025, *Fraunhofer AI Trans.*) ⁶.

¹ ² RIC-Net: A plant disease classification model based on the fusion of Inception and residual structure and embedded attention mechanism - ScienceDirect
<https://www.sciencedirect.com/science/article/abs/pii/S016816992100661X>

³ Lightweight Multiscale CNN Model for Wheat Disease Detection
<https://agris.fao.org/search/en/providers/122436/records/675aad2b0ce2cede71ccc836>

⁴ A comprehensive cotton leaf disease dataset for enhanced detection and classification - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S235234092400876X>

- 5 An attention enhanced CNN ensemble for interpretable and accurate cotton leaf disease classification | Scientific Reports

https://www.nature.com/articles/s41598-025-34713-w?error=cookies_not_supported&code=fa75afd7-b37a-44de-996f-d99a2c3ab9d7

- 6 COLD-12: A multi-level feature extraction hybrid CNN Model for accurate cotton disease diagnosis - ScienceDirect

<https://www.sciencedirect.com/science/article/pii/S2773186325000532>